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Applicability of Multi-Agent Simulation for Crowd Control After a Large-Scale Event

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ABSTRACT

This study aims to assess the applicability of multi-agent simulation for crowd control after large-scale events. It focuses on pedestrian pathways around a stadium in an urban area. By utilizing real-world pedestrian flow data, we investigated the parameters essential for simulation and analyzed crowd control. Consequently, we identified critical simulation parameters, such as speed–density relationships and personal distance, which produced highly accurate results across different days. In terms of congestion control, we established an acceptable congestion range based on data and illustrated that averaging peak bottleneck flows effectively alleviates congestion using the simulation.

1 | Introduction

At large-scale events such as sporting events and concerts, the simultaneous movement of people before the start and after the conclusion of the event leads to concentrated crowds. This congestion occurs not only within the facilities themselves but also in pedestrian pathways surrounding the facilities that connect to public transportation systems. The concentration of crowds carries the potential to lead to serious accidents.[1] Consequently, there is a pressing need to assess these risks and implement appropriate crowd control measures to mitigate them. However, much of the risk management in crowd security, such as crowd control, remains heavily reliant on the experience and expertise of seasoned professionals. This practice has yet to be sufficiently systematized from a scientific perspective, and there is an expectation that findings from scientific research will be incorporated [1].

In this context, multi-agent simulation (hereinafter referred to as MAS) [2] has emerged as a method recognized for its effectiveness in evaluating the safety and efficiency of crowd control. However, discussions on the use of MAS have primarily focused on simulations of evacuation planning for emergencies such as fires and earthquakes [3]. Applications of MAS to crowd control in large-scale events remain limited [4, 5], and there are few cases where safety has been evaluated based on data measured from actual event crowds.

Furthermore, MAS is characterized by its ability to model the behavior of the entire system through autonomous agents that can exhibit distinct behavioral patterns and decision-making rules. However, to conduct simulations that accurately reflect real-world conditions, it is essential to incorporate quantitative crowd characteristics related to speed and density into the agents' attributes. Many MAS-equipped simulators are designed for evacuation scenarios during fires and similar emergencies and cannot always be applied to the characteristics of everyday crowds in large-scale facilities that regularly host events.

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Furthermore, scientific understanding of crowd dynamics during post-event egress in large-scale events remains limited, and applications of MAS to such scenarios are scarce. To address this gap, this study aims to obtain knowledge for crowd control in regularly scheduled sports events through the following steps: (1) collecting pedestrian flow data in real-world environments, (2) validating MAS reproducibility by calibrating its parameters using the acquired data, and (3) performing a quantitative analysis of pedestrian flow control via MAS. The study focuses on pedestrian walkways around a stadium that regularly hosts large-scale sports events. The MAS framework employed in this study is implemented using the commercial software Pathfinder, which is described later in Section 6.

1.1 | Literature Review

1.1.1 | Multi-Agent Simulation and Pedestrian Dynamics Modeling

Multi-agent simulation (MAS) has emerged as a powerful computational approach for modeling complex crowd dynamics and pedestrian behavior. In that context, Sidiropoulos et al. [2] demonstrated that MAS-based approaches have been extensively applied to crisis management scenarios over the past decade, with demonstrated effectiveness in simulating evacuations and emergency responses. On the other hand, Helbing et al. [6] established foundational work on self-organized pedestrian crowd dynamics, building on their social force model [7], demonstrating that geometric boundary conditions (the physical shape and layout of spaces) significantly influence pedestrian flow characteristics. Their research on the social force model has become a cornerstone for understanding pedestrian movement patterns.

Several commercial and research-oriented simulation platforms have been developed to implement MAS algorithms. Pathfinder, buildingEXODUS, STEPS, MassMotion, and SimTread represent major crowd simulation software packages that use agent-based algorithms [8]. Thornton et al. [9] presented Pathfinder as an agent-based egress simulator that uses steering behaviors and physical constraints to simulate large-scale occupant behavior. Kuligowski [10] provided a comprehensive review of building evacuation models, noting that each platform varies in its approach to movement algorithms, behavior modeling, and fire effects incorporation. Validation studies have confirmed reasonable reproducibility through comparative verification with actual evacuation experiments [11], though Siyam et al. [3] noted that research issues remain regarding the realism of behavioral aspects in agent-based evacuation modeling.

1.1.2 | Crowd Management and Safety at Large-Scale Events

Crowd management at large-scale sporting events and public gatherings represents a critical safety challenge that has gained increasing attention following tragic incidents worldwide. Ha [12] conducted a systematic literature review on crowd stampede management at sporting events, emphasizing the need for a whole-society-based approach involving international organizations, regional governments, sports participants, and spectators. The review highlighted major incidents that underscore the pressing need for improved crowd control measures.

Weaver and Liu-Lastres [13] analyzed crowd risk mitigation technologies in urban sport events, noting that as more events draw large crowds, crowd management has become crucial for ensuring both public safety and optimal guest experience. They emphasized that proper crowd management requires understanding how technology, security, and human behavior interact in dynamic environments. Martella et al. [14] discussed the critical need for increased situation awareness, prediction, and intervention capabilities in current crowd management practices.

The flow characteristics and safety considerations for stadium ingress and egress have been extensively studied. Chin et al. [15] measured attendance, demographic distribution, egress times, and exit choice data in a contemporary Canadian stadium, using this empirical data to validate their social force model-based approach. Their research demonstrated the importance of comparing measured and simulated egress times for model validation. Ronchi and Nilsson [16] modeled total evacuation strategies for high-rise buildings in Pathfinder and cross-validated the results with STEPS (a fine-network model), comparing alternative strategies and identifying those that reduce total evacuation time. Larsson et al. [17] investigated the impact of crowd composition on egress performance, demonstrating that crowd composition, including factors such as body area occupied and social interactions, measurably impacts observed velocities, flows, and densities.

1.1.3 | Speed–Density Relationships and Fundamental Diagrams

Calibrating multi-agent simulations for post-event scenarios depends critically on accurate speed–density relationships. The fundamental diagram provides this foundation by describing how pedestrian speed, density, and flow relate to one another in various congestion conditions. Vanumu et al. [18] conducted a comprehensive review of fundamental diagrams of pedestrian flow characteristics, emphasizing that understanding these relationships is crucial for accurately estimating system capacity, particularly in transportation terminals where large crowds gather and transfer. They noted that fundamental diagrams play an important role in developing dynamic simulation models by providing equilibrium relationships between macroscopic variables.

Seyfried et al. [19] re-examined pedestrian fundamental diagrams from controlled corridor experiments and showed that the observed speed–density relationship is sensitive to how the experiment is set up and how the data are evaluated, so reliable capacity estimates require clearly defined measurement conditions. Later, Steffen and Seyfried [20] introduced a Voronoi-cell–based measurement method that assigns each pedestrian a personal space and computes density at that local scale; this reduced scatter in the data and made it possible to analyze speed–density relations and interaction effects at almost step-by-step resolution.

Kadali and Vedagiri [21] reviewed pedestrian level-of-service methods for facilities such as sidewalks, crosswalks, and other walkways, and showed that many of the commonly used LOS criteria were developed for conditions unlike those found in dense,

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mixed, or developing-country environments, so the thresholds often need local adjustment. The LOS idea itself goes back to Fruin [22], who tied space per person and flow (persons per meter per minute) to perceived comfort and defined LOS A–F for ordinary transport/shopping-type facilities, where operations were well below extreme crowding. Still [23] later documented that event and stadium crowds can operate at densities higher than those implicit in Fruin's tables, meaning that LOS values taken from Fruin [22] should be interpreted carefully for venue-type, purpose-driven crowds rather than applied as universal limits.

1.1.4 | Post-Event Egress and Stadium Crowd Flow

Post-event egress from large venues represents a unique challenge distinct from emergency evacuation scenarios. García et al. [24] analyzed pedestrian access to a football stadium and showed that fan arrivals at multiple gates can create complex, locally dense situations, so simulating pedestrian dynamics in the full, irregular geometry is necessary. They built a simulation framework that combines a social-force model with precomputed desired-velocity fields and validated it against actual turnstile entry-rate data from several matches; the calibrated model reproduced the time evolution of incoming crowds across those games, indicating that a single, data-informed parameterization can be reused for that venue.

Larsson et al. [17] analyzed egress from a large UK stadium for four different event types (football, rugby, concerts and an American football game) and showed that crowd composition together with external factors such as weather measurably affects the observed velocities, flows and densities. Importantly, all observed flow rates were below the maximum value of 82 persons/m/min recommended in the UK Guide to Safety at Sports Grounds [25], indicating that design flow values taken from guidance should be applied with caution when the expected crowd composition differs from the assumptions in the guide.

Still [23] argued that stadium egress based solely on handbook peak flow values leaves only a narrow operational margin, because actual event crowds can reach local densities higher than those observed in the transport- and retail-oriented datasets reported by Fruin. Drawing on observations at Wembley Stadium, they documented densities about 4.7 persons/m^2, that is, above the levels implicit in Fruin's LOS tables, and noted that under such conditions small disturbances can rapidly degrade movement. In these high-density regimes, the speed–density relationship cannot be treated as a single, smooth curve determined only by spacing; behavioral responses to pressure, local stopping, and compliance with on-site instructions become additional factors that must be considered.

1.2 | Research Gaps and Significance of Current Study

Although extensive literature exists on pedestrian simulation, the predominant focus remains on emergency evacuation rather than routine post-event dispersal from sports venues. Reviews of agent-based simulation approaches have identified a persistent bias toward crisis scenarios and noted ongoing challenges in behavioral calibration and validation [3, 10]. While empirical studies of crowd movement at venues exist, comprehensive stadium egress research that analyzes multiple ordinary events using measured data for both model calibration and validation remains limited. Where high-quality observational data is available, findings demonstrate that observed flow rates during post-event egress fall significantly below handbook design values and vary substantially with event type and environmental conditions [17]. This discrepancy underscores that evacuation-calibrated parameters serve as poor proxies for normal, non-emergency crowd behavior.

Validation methodology represents another gap in literature. Many comparative studies benchmark simulation models against controlled drills or laboratory experiments rather than live post-event crowds, with results showing strong sensitivity to configuration choices and user-defined parameters [11]. Furthermore, even when models are calibrated to specific events, the transferability of those parameters across different days or event types at the same venue is rarely demonstrated, despite its operational significance for venue management.

The majority of empirical speed–density relationships documented in the literature derive from simple geometric configurations such as straight corridors and single-exit facilities [19, 20]. The application of these relationships to complex real-world networks—incorporating multiple route choices, changes in elevation, and signal-controlled pedestrian crossings—and their translation into actionable operational targets for stewards and transport operators remain underdeveloped areas of research. Practitioners have explicitly identified the need for enhanced situation awareness, predictive capabilities, and intervention tools that can translate analytical findings into implementable crowd management strategies [14].

This research addresses the identified gaps through four principal contributions: (1) measurement of post-event pedestrian flows, densities, and walking speeds during routine operations at a regularly utilized stadium venue; (2) calibration of multi-agent simulation using empirically derived parameters rather than evacuation-scenario defaults; (3) validation of temporal agreement across independent event days using quantitative performance metrics; and (4) development of implementable operational control strategies based on simulation findings, specifically targeting sustainable throughput thresholds through peak-smoothing interventions and route guidance measures. Collectively, these contributions establish an evidence-based framework for post-event crowd management at large-scale venues, bridging the gap between pedestrian dynamics research and operational practice.

2 | Methodology Overview

In accordance with the research methodology outlined in Figure 1, this study conducts the following analyses:

1. Measurement of Pedestrian Throughput and Crowd Characteristics

   Cameras were installed along pedestrian egress routes to measure the number of passing individuals, walking speeds, and crowd densities during 3 days (June 1, 3, and 4, 2022). Data collection spanned 60 min before and 120 min after the match end time. The measurement methodology is detailed in Section 4.

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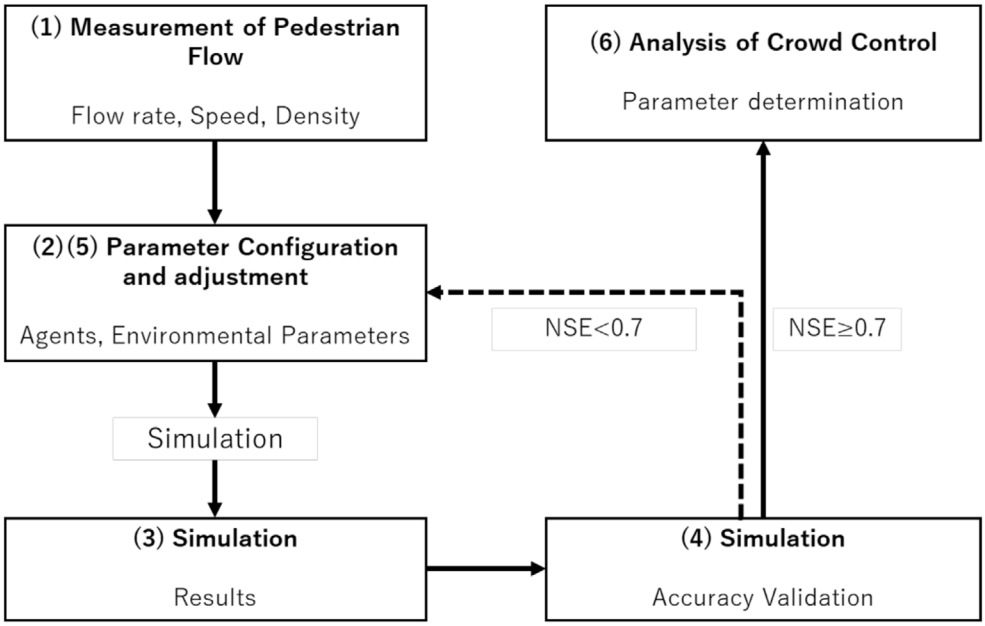


FIGURE 1 | Research procedure.

2. Parameter Configuration for Crowd Simulation

   The default parameters of the simulation software are configured for evacuation scenarios, necessitating adjustments to align with the normal post-event egress conditions examined in this study. To address this, parameters such as walking speed are set based on empirical data obtained from field measurements. Detailed descriptions of these configurations are provided in Sections 6.1–6.3.

3. Crowd Simulation

   Pedestrian counts measured at the starting point of the egress routes are used as the initial input for the simulation. Using the parameters configured in Step (2), crowd dynamics are simulated. Details are provided in Section 6.4.

4. Validation of Simulation Accuracy

   The discrepancy between measured and simulated pedestrian count is evaluated at sensor lines installed in congested zones. The Nash–Sutcliffe efficiency (NSE) [26, 27] is used as the validation metric. The methodology is detailed in Section 7.

5. Parameter Refinement

   To improve the evaluation values of step (4), the parameters are adjusted and steps (2) to (4) are repeated.

6. Analysis of Crowd Control

   Crowd control is examined based on the measured data and simulation results.

3 | Study Area

This study analyzes pedestrian flow in the walkways surrounding a dome facility located in an urban area, specifically immediately after the conclusion of a baseball game. In particular, the study focuses on the route leading from the dome facility, where congestion is consistently observed after games, to the west exit of Station A. Figure 2 illustrates the pedestrian routes under study and the measurement points for pedestrian data collection. There are two main routes taken by pedestrians heading toward the west exit of Station A. The first route (hereafter referred to as Route 1) follows the primary walkway, where pedestrians exit the dome facility, pass through measurement line L1 (hereafter L1), walk along the pedestrian deck level, cross a footbridge (measurement line L3, hereafter L3), and proceed via Crosswalk 2 to reach the west exit of Station A.

The second route (hereafter referred to as Route 2) follows a path where pedestrians descend the stairs from measurement line L2 (hereafter L2) to the ground level, proceed through Crosswalk 1, pass through measurement line L4 (hereafter L4), and continue via Crosswalk 2 to reach the west exit of Station A. Additionally, for Route 1, after passing through L1, some pedestrians may choose an alternative route (hereafter referred to as Route 3) to avoid congestion on the footbridge by taking a detour toward the east exit of the station. Although there are several other minor walkways that function as shortcuts, the number of pedestrians using them is lower compared to the two main routes mentioned above; therefore, they are excluded from consideration in this study.

The measurement dates and spectator numbers are summarized in Table 1. Measurements were conducted over 3 days—June 1, 3, and 4, 2022—with the match start time set at 18:00 on all days. The weather was either cloudy or clear. The number of spectators ranged from approximately 32 000 to 40 000, with restrictions on audience size due to COVID-19 having already been lifted. To analyze the crowd congestion caused by the simultaneous departure of spectators after the end of matches, pedestrian counts were conducted over a total of 180 min: 60 min

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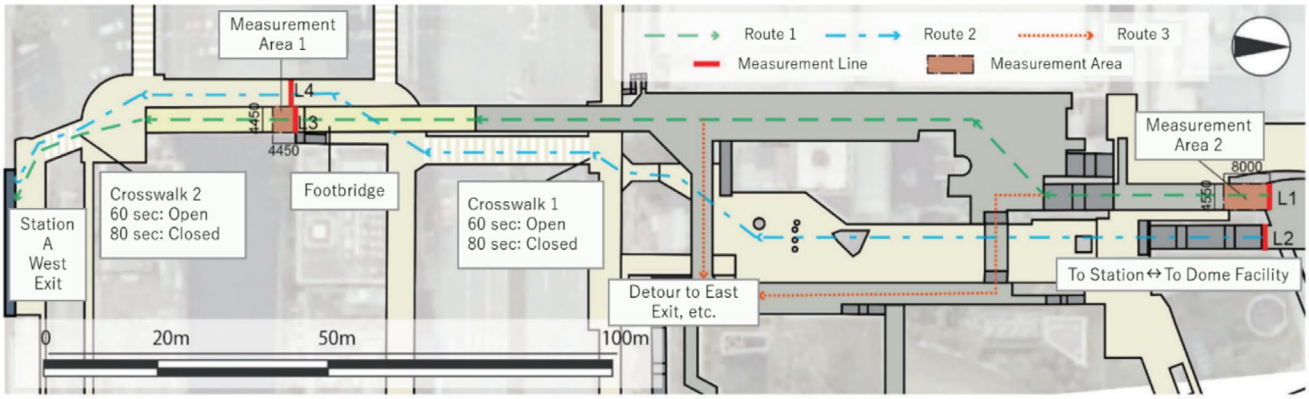


FIGURE 2 | Scope of research subject (The CAD data was created by the authors and overlaid on aerial photographs from the Geospatial Information Authority of Japan). None of the pedestrian measurements included attribute data; only macroscopic quantities were recorded.

TABLE 1 | Measurement dates and match information.

	Match start time	Match end time	Match duration	Number of spectators	Weather
June 1 (Wed)	18:00	20:44	2:44	32058	Cloudy
June 3 (Fri)	18:00	21:19	3:19	40392	Clear
June 4 (Sat)	18:00	20:21	2:21	40444	Clear

before and 120 min after the match end time, using the match end time as the reference point.

4 | Crowd Measurement Method

Crowd flow was measured by installing video cameras along pedestrian walkways and analyzing the recorded footage. For pedestrian counts, a measurement line (at a height of 1500 mm) was set on the monitor screen, and the number of heads crossing the line per minute was counted and recorded. Two observers conducted the counting, and the average of their counts was used for analysis.[2] To measure crowd density, Measurement Area 1 (4450 mm × 4450 mm, corresponding to the width of the footbridge) was set on the footbridge adjacent to L3. Still images were extracted from the video at 1-min intervals, and the number of heads within the area was manually counted to calculate the crowd density. Observer 1 marked each head on the still image, while Observer 2 cross-checked for overlapping pedestrians using the video and made corrections as needed. Walking speed was measured using a 10-min non-congested period starting 60 min before the end of the match on June 4. A total of 110 pedestrians were randomly sampled as they passed through Measurement Area 2 (4550 mm × 8000 mm).[3]

5 | Measurement Results

Table 2 shows the number of people passing through the measurement lines on each measurement day. It indicates the number of people passing through the lines during the 180-min period (hereinafter referred to as the entire range $t_{-60-120}$) starting 60 min before and ending 120 min after the game end time t_0 on each measurement day, as well as during the 60 min after t_0 (hereinafter, t_{0-60}), and the time from t_0 until the congestion on L3 ends, t_{ec} (hereinafter, t_{0-ec}). During $t_{-60-120}$, the number of people assumed to have headed toward Station A West Exit passing through L3 and L4 was about 26%–28% of the attendees, with about 58%–70% of them concentrated in the t_{0-60} period.

Figures 3–5 show the time-series changes in the number of people passing the measurement lines on each measurement day. Figures 6–8 show the relationship between the number of people passing through L3 and the crowd density (Measurement Area 1) during t_{0-60}. On all measurement days, a sharp increase in pedestrian flow increased sharply immediately after the game ended. About 5 min after the end, the maximum passage rate at L1 was measured at approximately 200–230 people/min, with a flow coefficient of about 44.9–51.7 people/m/min. Looking at L3, on all measurement days, from around 10 min after the game ended, large fluctuations in the passage rate were observed, ranging from 60 to 130 people/min (13.2–28.6 people/m/min), accompanied by a sharp increase in density. This is due to the stop-and-go congestion caused by Crosswalk 2 (Figure 2) located ahead of the pedestrian bridge, which occurred regularly. The start and end times of the period with regular congestion (hereinafter, steady-state period) are indicated by the range between the two black vertical lines in Figures 6–8. This range was manually determined by extending from the point of maximum density to where the trends in density and passage rate significantly change. The steady-state congestion is thought to occur because the combined flow of L3 and L4 exceeds the capacity of Crosswalk 2. Based on the 5-min moving average (hereinafter, MA5min) during the steady-state period, the approximate capacity of Crosswalk 2 before congestion occurs is estimated to be 160–170 people/min. The corresponding flow coefficient is about 22.9–24.3 people/m/min,[4] which is approximately 50% of the maximum passage rate at L1.

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TABLE 2 | Number of people passing through the measurement line.

Measurement date	Measurement period (min)		Passing count (persons) L1	L2	L3	L4	Number of entrants (persons)	x 1 − (L3/L1)
June 1	Entire	$t_{-60-120}$	6685	1603	6659	1870	32058	0.23
	60 min	t_{0-60}	4493	1392	4162	1329		
	Steady period	t_{0-28}	3732	1290	2871	1059		
June 3	Entire	$t_{-60-120}$	9167	1998	8701	2467		
	60 min	t_{0-60}	5645	1689	4565	1857	40392	0.29
	Steady period	t_{0-35}	4965	1640	3535	1689		
June 4	Entire	$t_{-60-120}$	8757	2476	7642	2627		
	60 min	t_{0-60}	6979	2256	4992	2210	40444	0.35
	Steady period	t_{0-43}	6379	2225	4162	2057		

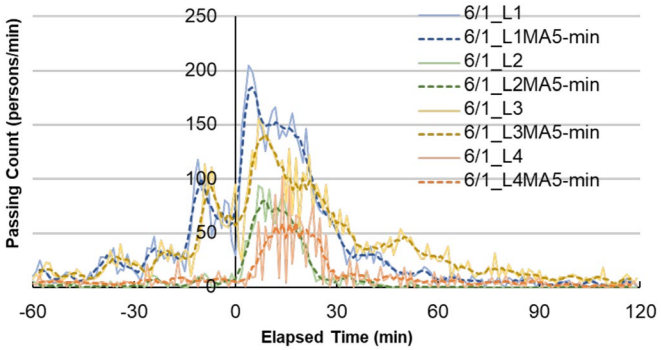


FIGURE 3 | Time series change in number of people passing through measurement lines (June 1).

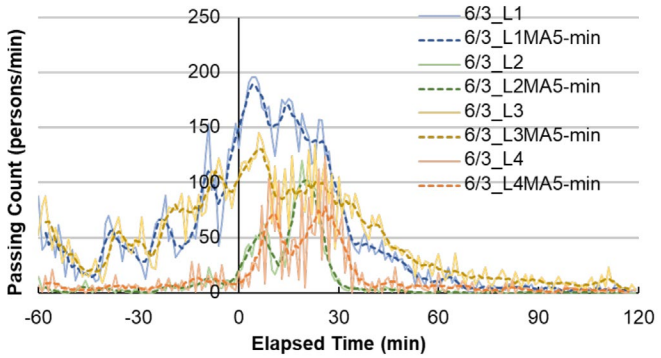


FIGURE 4 | Time-series change in pedestrian count at measurement lines (June 3).

Next, the video footage confirmed that congestion occurred on the footbridge during the steady interval, and that on each measurement day, confirmed that there was a density at which walking completely stopped (hereinafter referred to as maximum density) (as indicated by the dashed ellipses in Figure 9).[5] The average density during the steady interval was 1.23 ± 0.27 persons/m^2, and the Level of Service [21] (hereinafter referred to as LOS)[6] was primarily Level E. The average of the maximum densities at which walking came to a full stop was 1.87 persons/m^2. While this does not represent an immediately dangerous situation, given that walking was restricted and that the congestion occurred on a footbridge suggests that mitigation measures should be considered.

Regarding walking speed, the video analysis software Tracker was used to measure the walking speed of pedestrians who passed through L1 in Measurement Area 2. The result was 1.36 ± 0.21 m/s.[7] Although pedestrian walking speed is known to vary depending on the subject and environment, Pan et al. [28] reported 1.57 ± 0.12 m/s, and Ren et al. [29] reported 1.55 ± 0.13 m/s based on their experimental results. However, since walking speed tends to increase until about age 20 and then

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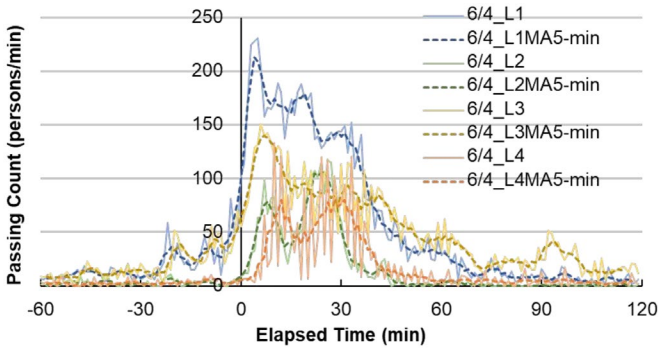


FIGURE 5 | Time-series change in pedestrian count at measurement lines (June 4).

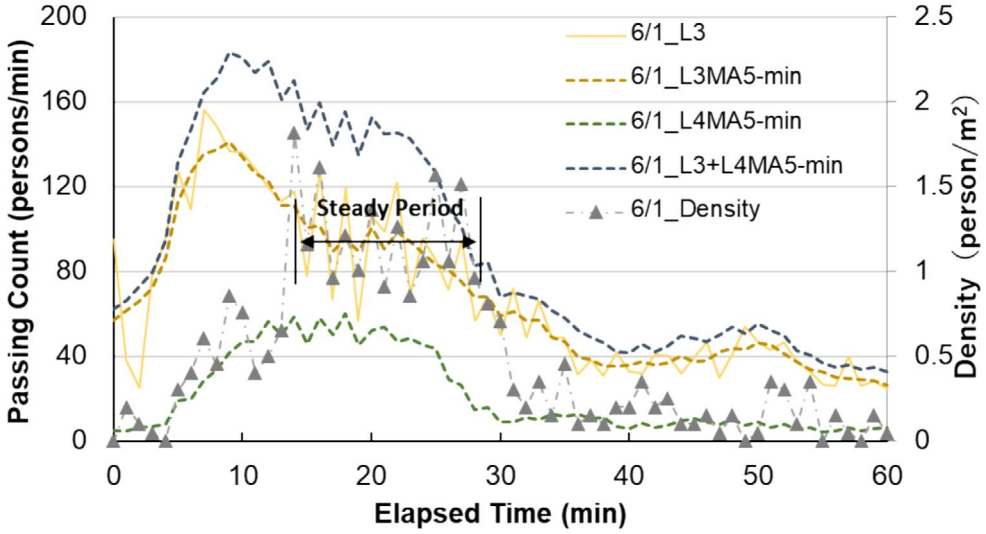


FIGURE 6 | Number of people passing through L3 and crowd density during t_{0-60} (June 1).

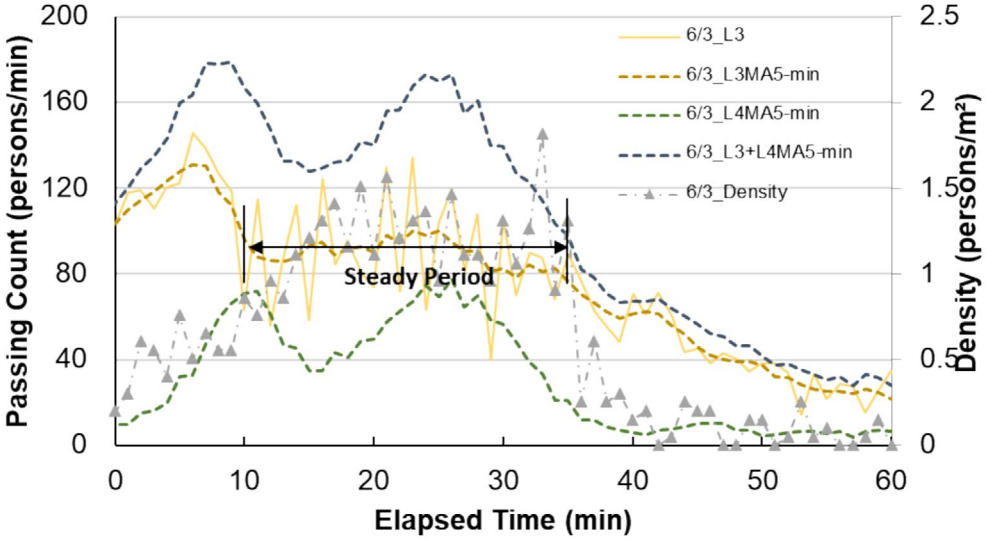


FIGURE 7 | Pedestrian count and crowd density at L3 during t_{0-60} (June 3).

decrease thereafter, and because the subjects in those experiments were all university students, it is assumed that the walking speed of a general crowd with diverse attributes would be slower. Meanwhile, Okada et al. [30] reported a normal walking speed of 1.17–1.25 m/s, and the average walking speed of elderly individuals is reported to be 0.94 m/s. Weidmann [31], based on an analysis of 12 previous studies, reported 1.34 ± 0.26 m/s, which is close to the result obtained in this study.

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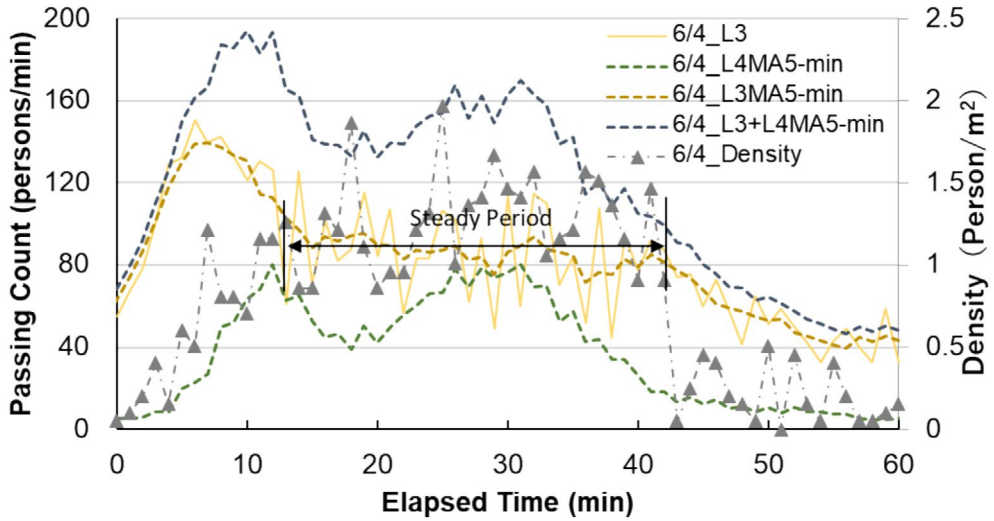


FIGURE 8 | Pedestrian count and crowd density at L3 during t_{0-60} (June 4).

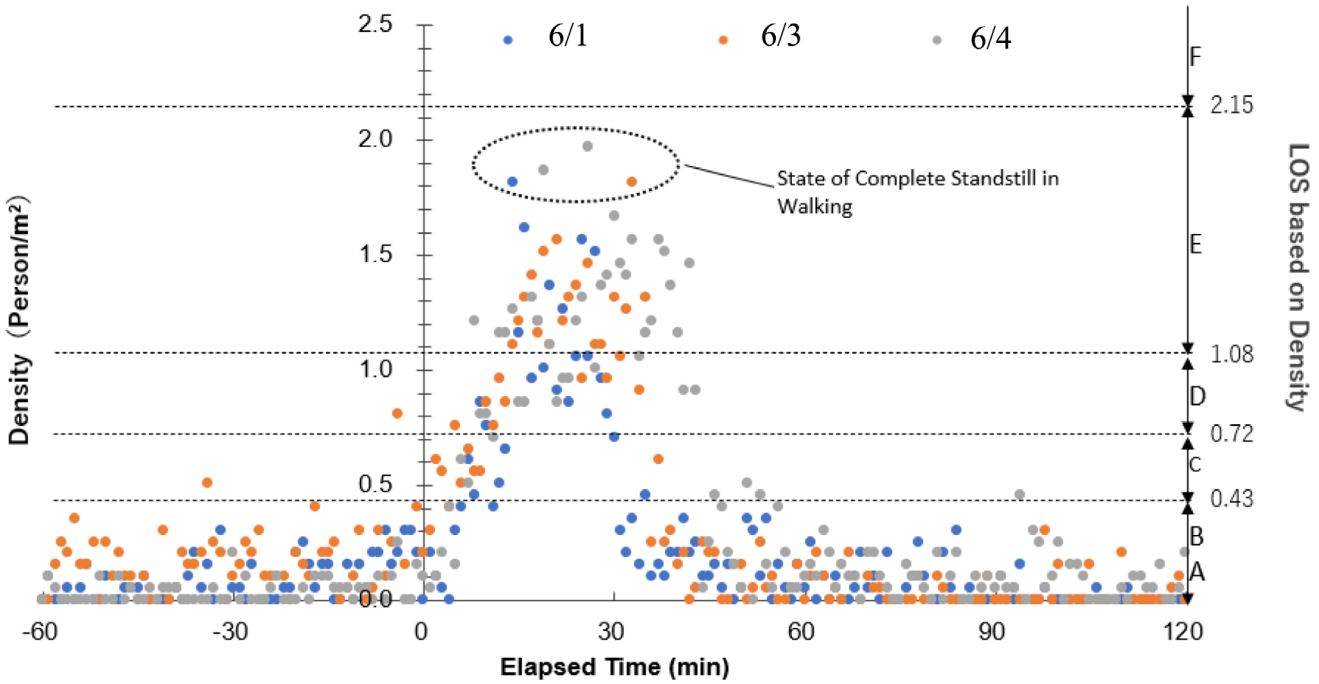


FIGURE 9 | Time-series change in crowd density.

6 | Simulation

6.1 | Overview of the Simulation Software

There are many crowd simulation environments that use MAS algorithms[8] [32–36], and a certain level of reproducibility has been confirmed through comparative verification with actual evacuation experiments [11]. In this study, to adequately reflect the characteristics of pedestrians observed under normal conditions into the simulation, Pathfinder 2022 was selected for the following reasons: the ability to link agent walking speed settings to density, the ability to control crowd density by specifying the distance between agents, the broad range of agent behavior control options, and the capability to model complex geometries in order to replicate real-world spaces.

Pathfinder models structures using a navigation mesh composed of triangular surfaces. These navigation meshes are grouped by rooms (Room) and connected via doors (Door) and stairs (Stair). Agents (Occupants) that move within the navigation mesh can be assigned attributes through profiles, which allow the configuration of characteristics such as body dimensions (Shape) and movement speed (Speed) (Figure 10). In addition, behaviors can be used to give agents various instructions (such as moving toward a destination, waiting, or changing behavior), and by combining different profiles and behaviors, agent groups can be classified in complex ways [36].

Here, only the profile parameters adjusted for this experiment are introduced; details on other parameters can be found in the Pathfinder user manual in [36].

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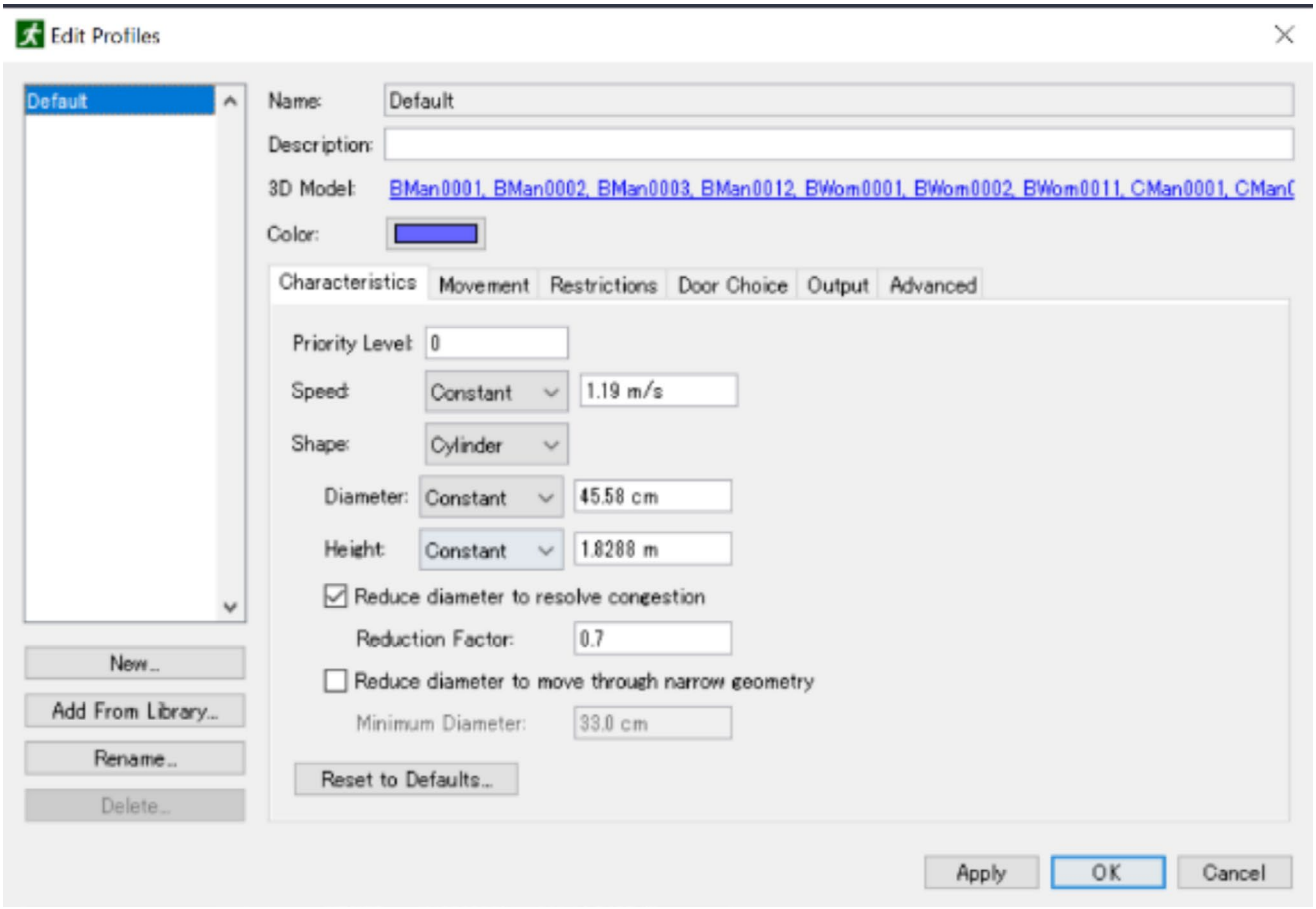


FIGURE 10 | Profile editing window in Pathfinder.

6.2 | Profile Settings

6.2.1 | Walking Speed (Speed)

In Pathfinder, the speed at which an agent moves on a horizontal surface is based on the walking speed and door flow rates referenced from SFPE [37], and is defined by Equation (1). The maximum speed v_{max} is the value entered under Speed in Figure 10, with a default value of 1.19 m/s. The coefficient v_{ft} accounts for the influence of the terrain on the agent's movement and allows for the effects of stairs or slopes to be considered (Equation 2). The default values for horizontal and sloped surfaces are $k = 1.4$ and $v_{ft} = 1.0$, respectively. In addition, $v_f(D)$ is a weighting factor used to adjust walking speed based on crowd density, as defined in Equation (3). Here, D is the crowd density (persons/m^2) in the room where the agent is located, and v_{fmin} denotes the minimum speed coefficient (default value: 0.15), as shown in Figure 11.

$$v = v_{max} \times v_f(D) \times v_{ft} \tag{1}$$

$$v_{ft} = \frac{k}{1.4} \tag{2}$$

$$v_f(D) = \begin{cases} 1, & D < 0.55 \text{ pers/m}^2 \\ \max\left[v_{fmin}, \frac{1}{0.85}(1 - 0.266D)\right], & D \geq 0.55 \text{ pers/m}^2 \end{cases} \tag{3}$$

The default walking speed parameters in Pathfinder are calibrated for evacuation behavior and therefore differ from the conditions assumed in this study, which focuses on typical situations such as returning home after an event. For this reason, in the present study, simulations were conducted using multiple models by adjusting the value of $v_f(D)$, with the crowd's average measured walking speed set to $v_{max} = 1.36 \pm 0.21$ m/s. Figure 11 shows the relationship between $v_f(D)$ and density examined in this experiment. Based on an initial value of $v_{fmin} = 0.15$, the density at which v_f reaches this minimum value [hereinafter $v_{fmin}(D)$] was varied within the range of 1.6–3.2 persons/m^2 for the simulation.

6.2.2 | Personal Distance

Personal distance (hereinafter PD) refers to the distance an agent attempts to maintain from other agents. It can be defined based on distance, density, or occupied area. In this experiment, PD $= c = 0.33$ m was derived from the average crowd density of 1.87 persons/m^2 measured on the footbridge under conditions of complete pedestrian standstill, and is given by Equations (4) and (5). The variable d represents shoulder width (diameter), and in this study, the initial value used is $d = 45.58$ cm.

$$c = \frac{2}{\sqrt[4]{12}}\sqrt{a} - d \tag{4}$$

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$$a = \frac{1}{\rho} \tag{5}$$

6.3 | Creation of the Navigation Mesh and Configuration of Crosswalk Open/Close Timing

The navigation mesh used in the simulation was created by extracting the 3D network structure of pedestrian pathways from Google Maps data publicly available on the internet and then verifying and correcting pathway widths and other details through on-site surveys to build a refined 3D model [38] (Figure 2). In Pathfinder, environmental elements other than the navigation mesh—such as door opening and closing—can also be configured. To construct a more realistic simulation environment, the timing of signal changes at Crosswalks 1 and 2 was measured and set to switch between 60 s open and 80 s closed as the initial values. Notably, Crosswalk 2 is located at the convergence point of Route 1 and Route 2 and acts as a bottleneck for the pedestrian flow heading toward the west exit of Station A, so its throughput is expected to influence the simulation results. On the other hand, because the road ahead is not very wide, pedestrians were occasionally observed ignoring the signal and crossing regardless. To account for this behavior and correct for crosswalk throughput, simulations were conducted with three open/close timing patterns for Crosswalk 2: 60–80 s, 62–78 s, and 64–76 s.

6.4 | Simulation Data and Model

The pedestrian inflow input data for the simulation used the measured values at L1 and L2 in Figure 2, and the simulation results at L3 and L4 were compared with measured values. Table 3 shows the types of agents used in the simulation. The input flow L_n consists of 1-min interval time-series data at each measurement line. x represents the proportion of people who, after passing through L1, detour to avoid congestion on the footbridge and choose Route 3. Here, x is calculated from the ratio between L1 and L3 during the period from the match end time t_0 to the end of congestion t_{ec} (Table 2).

In this study, after estimating the parameters v_{max}, c, and the detour ratio x from the measurement results, models were constructed by combining profile parameters such as $v_f(D)$ and environmental parameters such as the open/close timing of Crosswalk 2, and simulations were carried out.

7 | Simulation Accuracy Verification

Simulation accuracy was evaluated over the entire 180-min period $t_{-60-120}$, the 60 min after the match t_{0-60}, and the period until the end of congestion at L3 t_{0-ec}. The Nash–Sutcliffe

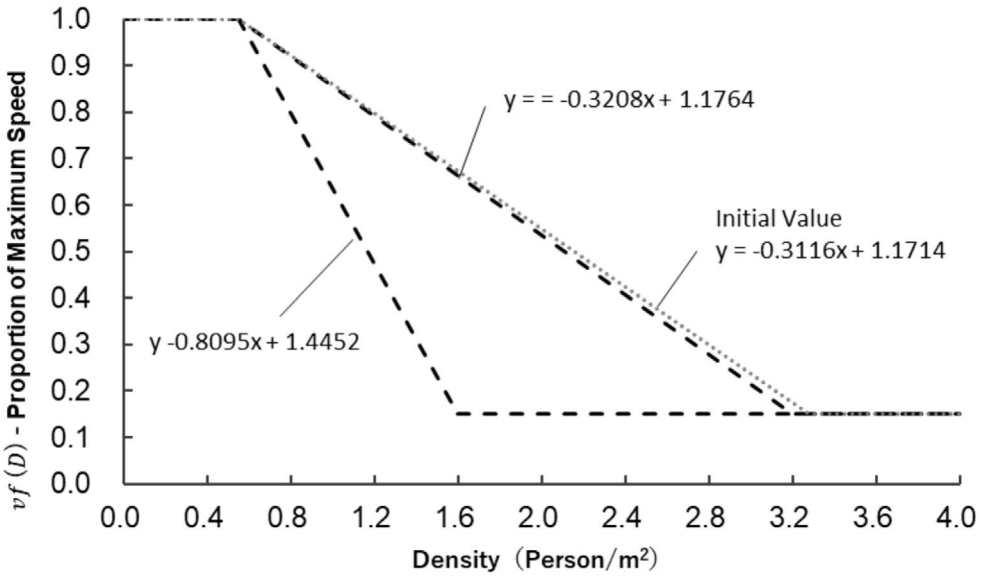


FIGURE 11 | Relationship between $v_f(D)$ and density.

TABLE 3 | Agent type.

Agent type	Profile		PD	Behavior	Inflow rate		
	Speed-density variable				$t_{-60} \le t < t_0$	$t_0 \le t < t_{ec}$	$t_{ec} \le t < t_{120}$
I	v_{max}	$v_f(D)$	c	Route 1	$L_1 t$	$L_1 t(1-x)$	$L_1 t$
II	v_{max}	$v_f(D)$	c	Route 3	—	$L_1 tx$	—
III	v_{max}	$v_f(D)$	c	Route 2	$L_2 t$	$L_2 t$	$L_2 t$

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coefficient (hereinafter, NSE) was used. NSE is a statistical index widely used in fields such as hydrology. It evaluates the prediction accuracy of the model by accounting for the magnitude of fluctuations in flow, where a value closer to 1 indicates higher accuracy. NSE is given by Equation (6), and in general, a value of 0.7 or higher is considered to indicate high model reproducibility [39].

$$\mathrm{NSE} = 1 - \frac{\sum_{t=1}^{T} \{(R(t) - S(t))\}^2}{\sum_{t=1}^{T} \left\{\left(R(t) - \overline{R}\right)\right\}^2} \tag{6}$$

Here, $R(t)$ refers to the result obtained in the real space, and $S(t)$ refers to the result obtained from the simulation. $\overline{R}$ is the overall average of the result $R(t)$ obtained in the real space.

8 | Simulation Results

8.1 | Examination of $v_f(D)$

The simulation parameter study was conducted using June 4,[9] the day on which the highest crowd concentration was expected. First, with the open/close time of Crosswalk 2 fixed at 60–80 s, simulations were conducted 10 times,[10] incrementally increasing $v_{\mathrm{fmin}}(D)$ from 1.6 to 3.2 in steps of 0.4.

Figure 12 shows the MA5-minute average NSE at L3 and L4 for each model. On June 4, the t_{ec} was 43 min. At L4, the NSE tends to increase as $v_{\mathrm{fmin}}(D)$ decreases. On the other hand, for L3, model 2.0 yielded the highest NSE values for $t_{-60-120}$ ("×" symbol) and t_{0-60} ("+" symbol) periods, and for t_{0-43} ("◆" symbol), model 2.4 yielded the highest value. From this, it is inferred that for the steady period at L3, the range $v_{\mathrm{fmin}}(D) = 2.0 - 2.4$ offers high reproducibility.

8.2 | Simulation Parameter Study Considering Crosswalk 2

Next, considering the reproducibility of the steady period, three patterns for the open/close time of Crosswalk 2 (60–80, 62–78, and 64–76 s) and three variations for $v^{(f_{\mathrm{min}})}(D)$ (2.0, 2.2, and 2.4) were changed, and nine models combining the two parameters were each simulated 10 times (Table 4).

Figure 13 shows the average NSE of the 5-min moving average for each model. Models E and F were able to achieve an accuracy of NSE > 0.7 in all periods.

Looking at the model characteristics by the open/close time of Crosswalk 2, when the opening time becomes longer, reproducibility improves in the t^{0-60} period for L3 ("+" symbol in the figure), and there is a tendency for reproducibility in the steady period t^{0-43} ("◆" symbol in the figure) to decrease. On the other hand, looking at the start and end times of the steady period for each model, the start time is nearly the same for all models, but the end time becomes earlier as the opening time of Crosswalk 2 gets longer, and the steady period becomes closer to the measured value (Figure 14). From these points, it is considered that for models A, B, and C—because the steady period is longer than the measured value—the reproducibility of the t_{0-60} period ("+" symbol) is low; for models G, H, and I—although the length of the steady period is close to the measured value—because the time-series trend of the pedestrian count differs, the reproducibility of the t_{0-43} period ("◆" symbol) is low; and it is estimated that models E and F have intermediate characteristics.

Next, looking at the model characteristics by the value of $v^{(f_{\mathrm{min}})}(D)$ similar to Figure 12, there is a tendency for the NSE value at L4 to increase as $v^{(f_{\mathrm{min}})}(D)$ becomes smaller. On the other hand, for

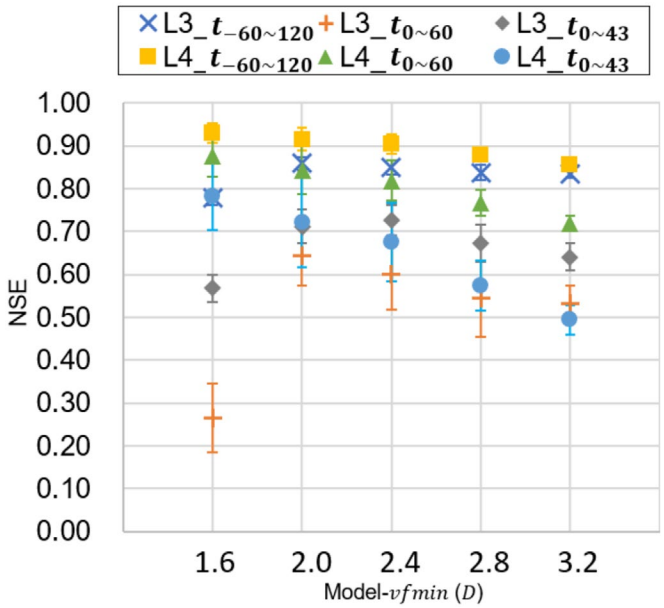


FIGURE 12 | NSE_MA5min—Crosswalk 2 open/close timing 60–80 s (June 4).

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the NSE at L3, it is at its maximum at 2.2 for the $t_{-60-120}$ ("×" symbol) and t_{0-60} ("+" symbol) periods, and it is about the same for 2.2 and 2.4 for the t_{0-43} ("◆" symbol) period. From this, it can be said that Model E is relatively more reproducible compared to others.

Furthermore, Figure 15 shows the maximum, minimum, median, and average values of the 1-min pedestrian count for t_{0-43}, which includes the steady period. For all of the models, the median and average values do not change much from the measured values, but there is a tendency for the maximum value to be larger than the measured value and the minimum value to be smaller than the measured value.

8.3 | Analysis of the Optimal Parameter Model

Next, using Model E—which showed the highest reproducibility in this experiment—we analyze the time-series changes in the number of people passing through L3 and L4 and in density, based on 1 of the 10 simulation runs where the sum of the NSE values was at the median.[11]

TABLE 4 | Types of simulation models.

		Crosswalk 2 open/close timing (seconds)		
		60–80	**62–78**	**64–76**
$v_{\text{fmin}}(D)$	2.0	A	D	G
	2.2	B	E	H
	2.4	C	F	I

Figure 16 shows the time-series changes in cumulative pedestrian counts at L3 and L4, where *S* indicates the simulation result. Both L3 and L4 exhibit similar trends overall, and even in the steady period, the slope of the graph is almost identical.

Figure 17 shows the time-series changes in pedestrian counts at L3. In the 60 min before the match ended, the waveform of the simulation results closely matches that of the measured data. This suggests that the hypothesis—that most visitors passing through L1 before the match ended continued toward L3—is generally valid.[12] On the other hand, the waveform after 60 min post-match differs significantly, likely due to inflow from a martial arts event held at another facility. That event ended at 9:40 p.m. (79 min after the match ended), and around that time, the flow rate increased. In Figure 16, we also see a large divergence in cumulative counts beginning around 60 min after the match ended. Since the shortest return route from the other facility to L3 does not require passing through L1, this led to a discrepancy with the simulation, which assumes passage through L1.

Looking at the MA5-minute average, the flow from the post-match peak to the end of the steady period was mostly reproduced. However, in the 1-min waveform, the measured data showed smaller fluctuations than the simulation, and the density was concentrated within a narrower range (Figure 18). As seen in Figure 15, for all models, the median and average values were close to the measured data, but the maximum–minimum range was wider than in the actual data. This suggests that the characteristics of movement in the simulation differ from real-world crowd behavior during the steady period. In both cases, stop-and-go crowd waves were observed, but in the simulation, their impact appeared more strongly. Once the agent ahead starts moving and space opens up, the agents in the simulation

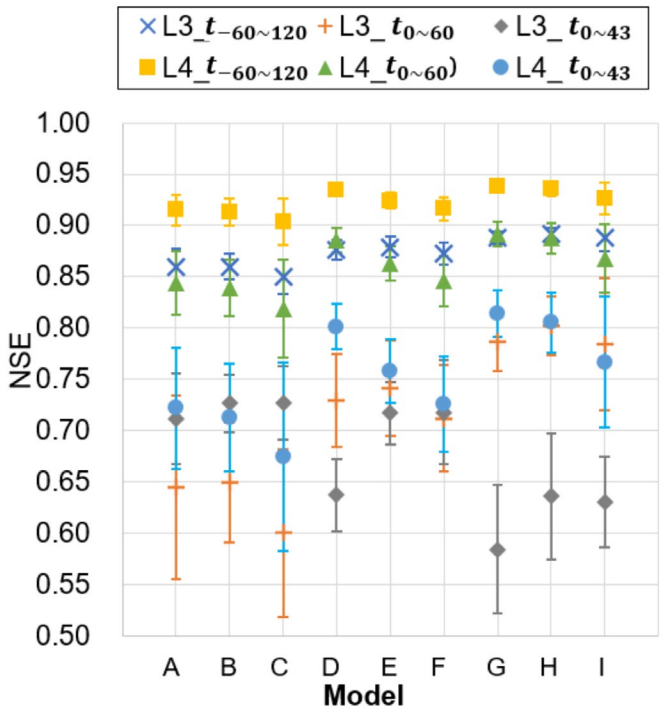


FIGURE 13 | NSE_MA5-min (June 4).

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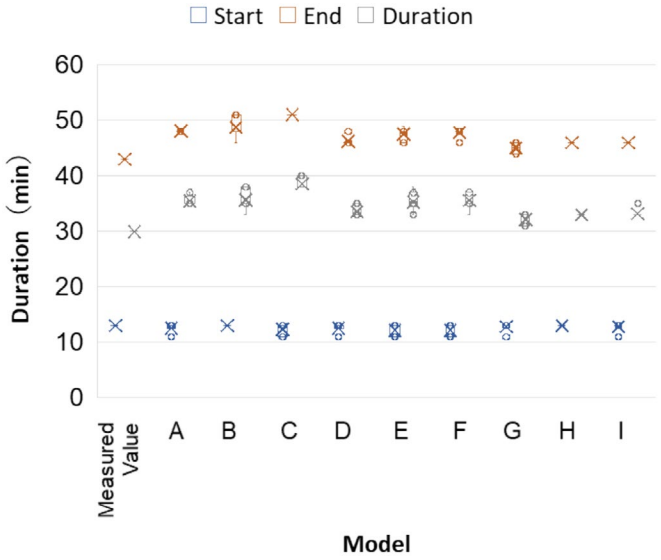


FIGURE 14 | Start and end time of the steady period (June 4).

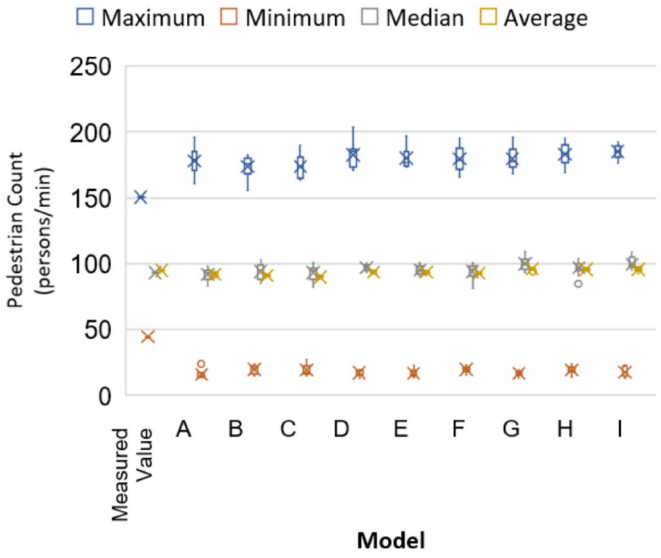


FIGURE 15 | t_{0-43} pedestrian count (June 4).

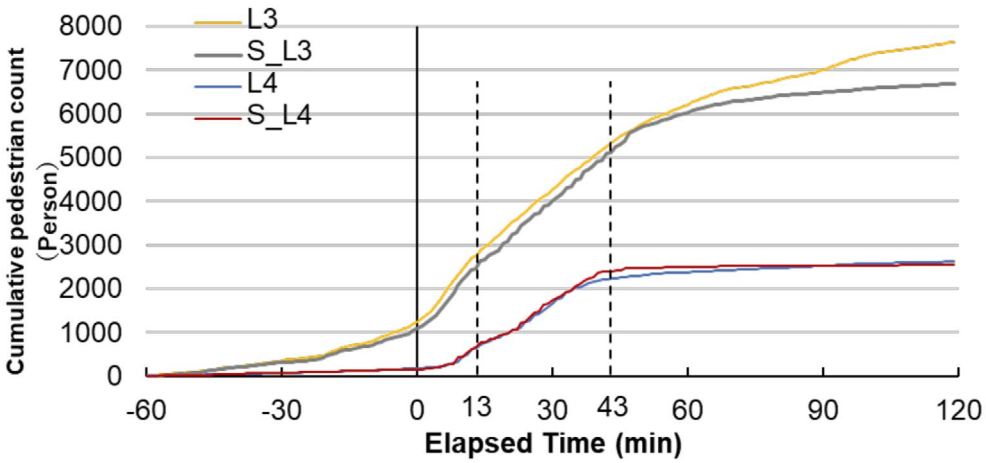


FIGURE 16 | Time-series change in cumulative pedestrian count (June 4)—Model E.

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begin to move more slowly than actual pedestrians, resulting in larger spacing fluctuations between dense and sparse regions.

Next, Figure 19 shows the time-series change in pedestrian counts at L4. The MA5-minute average overall was well reproduced, especially until the first post-match peak, which closely matched measured data. However, during the second peak, a small steady period appeared in the simulation that was not observed in the measured data. Similar to the case of L3, this is considered to be due to a simulation-specific characteristic: after the agent ahead starts moving and a gap appears, the following agents begin to move more slowly than actual pedestrians,

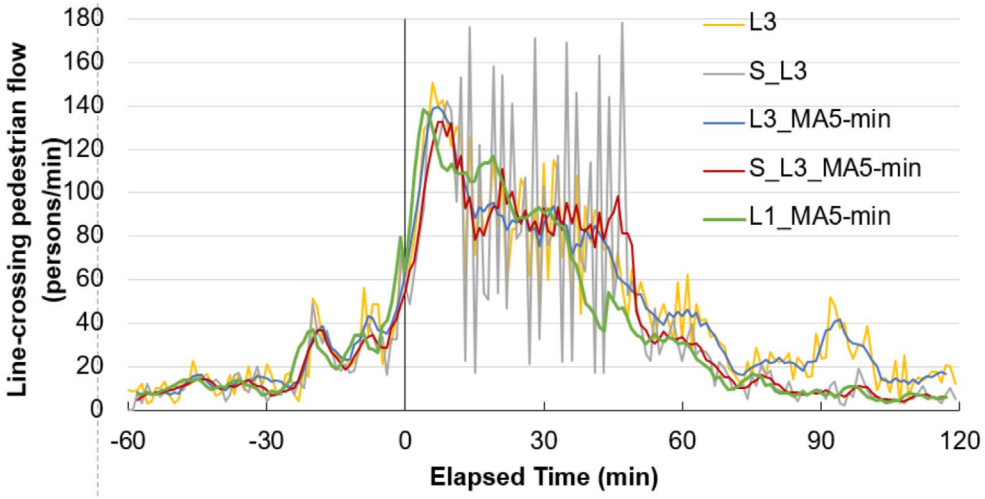


FIGURE 17 | Time-series change in pedestrian count at L3 (June 4)—Model E.

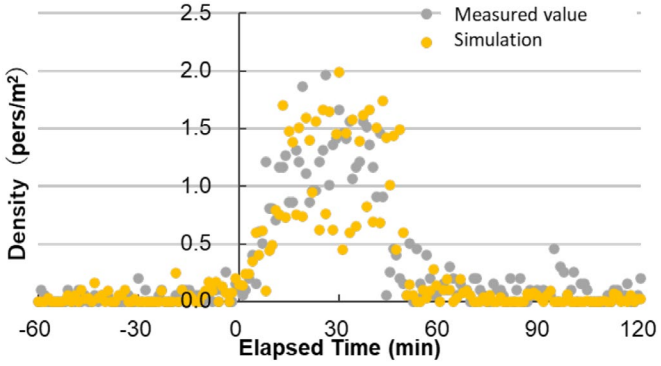


FIGURE 18 | Time-series change in crowd density (June 4)—Model E.

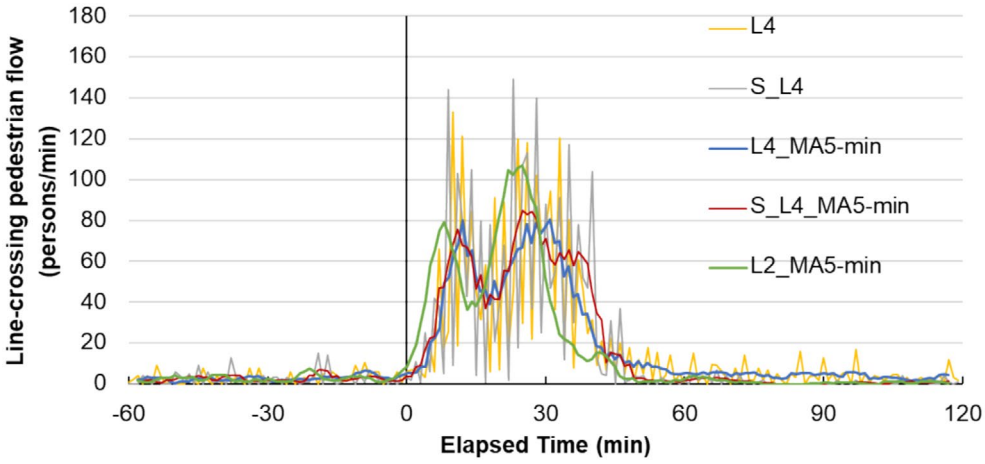


FIGURE 19 | Time-series change in pedestrian count at L4 (June 4)—Model E.

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causing larger undulation in density and increasing the likelihood of congestion.

8.4 | Application of Parameters to Other Days

To verify reproducibility under the same parameter settings as Model E, simulations were conducted on June 1 and June 3. The value of $v_{\mathrm{fmin}}(D)$ was set to 2.2, and the open/close time of Crosswalk 2 was set to 62–78 s. The end of congestion t_{ec} was assumed to be 28 min for June 1 and 35 min for June 3. Figure 20 shows the MA5-minute average NSE for both days. Looking at Model 0601, although the overall NSE was above 0.7, the value for L4 during the $t_{0-\mathrm{ec}}$ period ("●" symbol in the figure) was 0.58, which was relatively low. In Figure 21, the value of S_L4_MA5 is much larger than L4_MA5, suggesting that the assumed flow for people moving from L2 to L4 along Route 2 was incorrect.

Simulating under the assumption that all people passing through L2 head to L4 resulted in discrepancies. It was considered that on June 1, some people were heading elsewhere. Thus, it was hypothesized that during the steady period, 18% of those who passed through L2 went toward the east exit of the station, and 72% went toward the west exit.[13] A simulation based on this hypothesis was labeled Model 0601_op. In this model, the NSE values for all time segments improved.

For Model 0603, all NSE values except for L3 during $t_{0-\mathrm{ec}}$ ("◆" symbol in the figure) were higher than 0.79. According to Figure 22, the S_L3_MA5 values during the steady period were consistently higher than L3_MA5, suggesting that the second peak seen in the simulation was absent in actual data. This implies that congestion from L3 toward Station A was less severe in reality.

Therefore, it was assumed that some of the people who chose Route 3 to avoid footbridge congestion after passing L1 were actually heading to L4. Their proportion was assumed to be 3%,[14] and the resulting simulation was labeled Model 0603_op. Compared to Model 0603, the NSE values in Model 0603_op improved across all segments, though L3 during $t_{0-\mathrm{ec}}$ ("◆" symbol) still reached only 0.67.

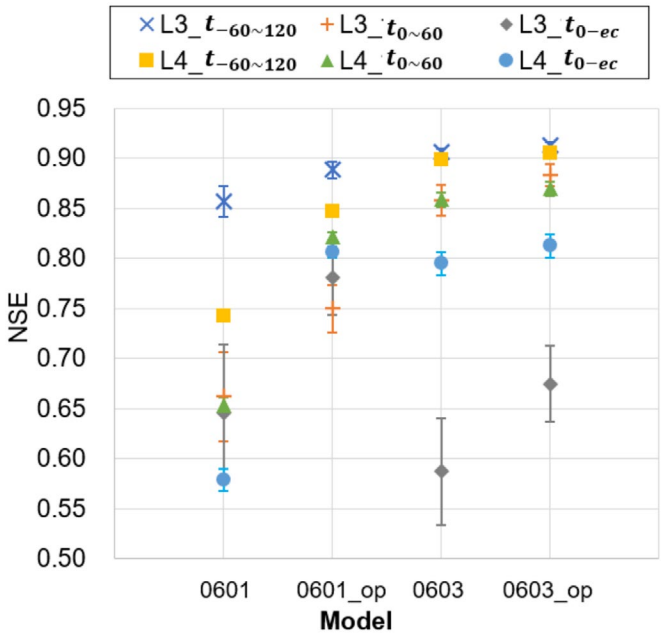


FIGURE 20 | NSE_MA5min (June 1 and June 3).

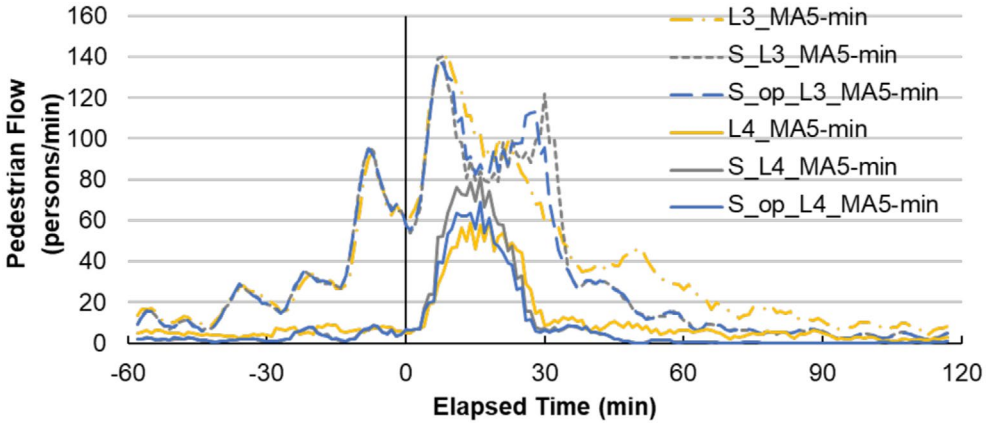


FIGURE 21 | Time-series change in pedestrian count at L3 and L4 (June 1).

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One reason for this mismatch may be the influence of another event conducted nearby. Especially on June 3, the ending time of that event (9:00 p.m., or 19 min before the match ended) was very close, suggesting that it affected congestion near the steady period.

On both days, applying the same parameters as June 4 allowed high reproducibility overall, but to further improve reproducibility during t_{0-ec}, adjustments in pedestrian flow assumptions are needed. Two reasons for this are proposed: first, differences in commuting patterns across days mean route choice proportions and patterns vary; second, match end times sometimes coincide with other nearby events' end times, affecting return flows.

8.5 | Crowd Control for Congestion Mitigation

Using Model E from June 4, we explore measures for mitigating congestion on the footbridge. Based on measurement results, it is estimated that congestion does not occur at Crosswalk 2 when flow is at or below 160–170 people per minute. Therefore, a simulation was run to test whether congestion could be eased by flattening the peak flow at L1 and L2 without changing the total flow.

As shown in Figure 23, the inflow to L1 and L2 during t_{0-43} was evenly distributed without changing the total number of pedestrians. Figure 24 shows the number of pedestrians and crowd density at L3 and L4. The peak density stayed below 0.8 people/m^2, and no congestion occurred on the footbridge during the steady period.

In reality, such precise control of pedestrian inflow as in the simulation is unlikely. However, by dispersing crowds using signage or guidance and flattening peak inflows onto the footbridge, it may be possible to ease congestion without reducing total pedestrian volume. One such method is to promote detouring through controlled crowd guidance, but doing so may cause new congestion elsewhere, therefore further investigation is needed.

9 | Conclusion

This study focused on pedestrian pathways surrounding a dome facility where large-scale sporting events are regularly

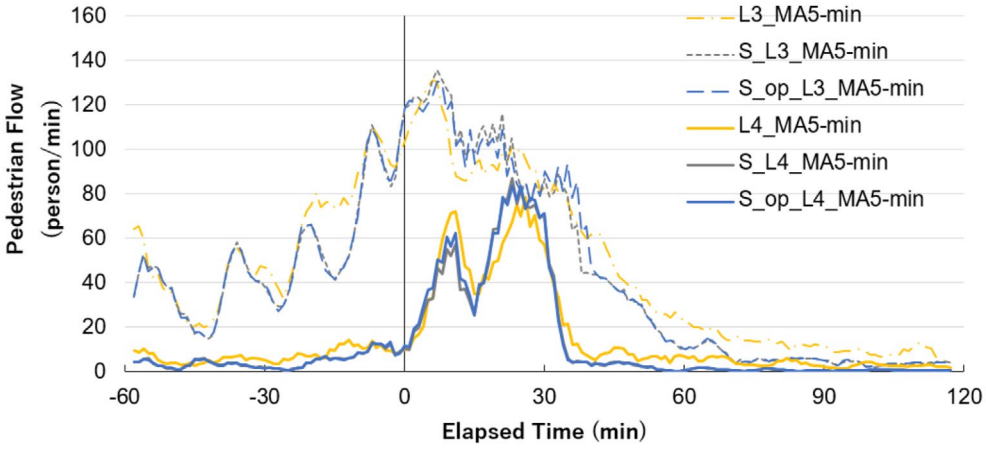


FIGURE 22 | Time-series change in pedestrian count at L3 and L4 (June 3).

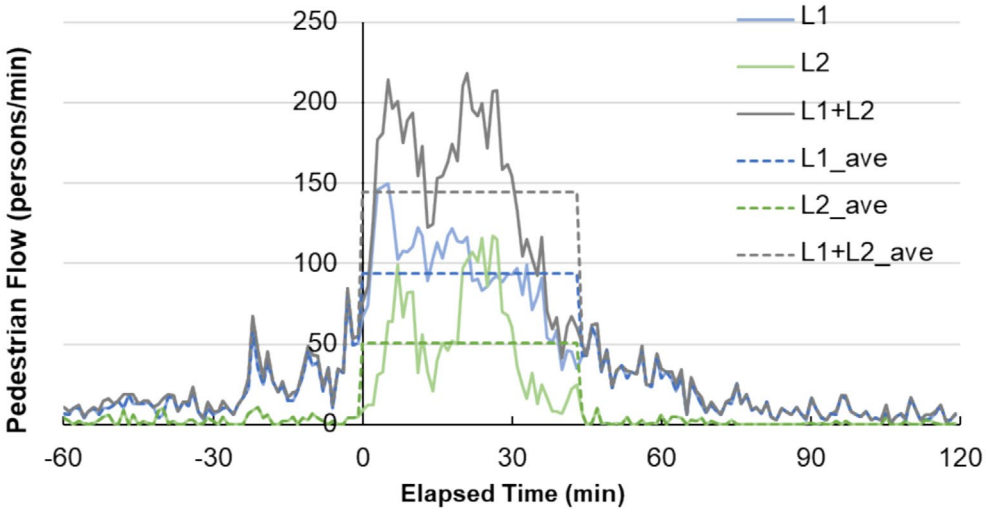


FIGURE 23 | Pedestrian flow at L1 and L2 (June 4).

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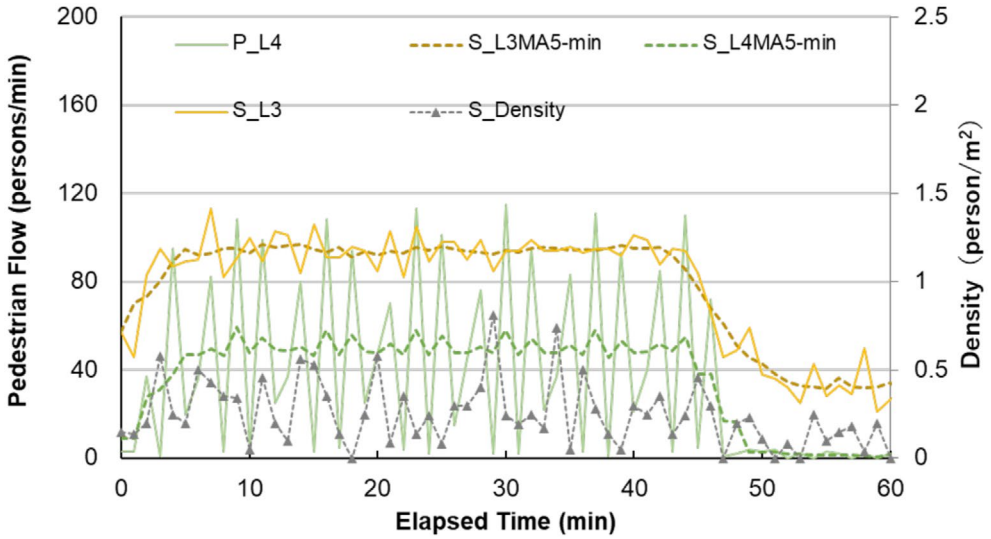


FIGURE 24 | Pedestrian count and crowd density at L3 and L4 (June 4).

TABLE 5 | Match progress [40–42].

		1	2	3	4	5	6	7	8	9	Total
6/1	Visitor	0	0	0	3	3	0	0	0	0	6
	Home	0	0	0	1	0	1	1	0	0	3
6/3	Visitor	0	0	0	0	0	0	0	3	0	3
	Home	0	1	2	1	1	0	0	5	×	10
6/4	Visitor	0	0	0	0	1	0	0	0	0	1
	Home	0	0	0	0	0	0	2	0	×	2

held. Using real-world pedestrian flow data, we examined the simulation parameters required and discussed crowd control. As a result, we identified parameters necessary for simulation such as the relationship between speed and density and personal distance. We also confirmed that simulation accuracy with NSE ≥ 0.7 could be achieved. Furthermore, we verified that the same parameters could be applied to different days.

Regarding crowd control, we clarified the congestion threshold values at bottlenecks based on measurement data and found that congestion could be mitigated by flattening the peak of inflow into the bottleneck. However, this study focused on only a portion of the pedestrian pathways around the dome facility, and did not consider the impact on other routes taken by detouring pedestrians. Therefore, a comprehensive study covering the entire pedestrian pathways is necessary. Additionally, since the acceptable capacity of Crosswalk 2 was roughly estimated within the study area, we believe that could actually be applied to crowd control by observing and measuring passage counts at L1 and L2 or at locations closer to the dome facility. However, another future challenge will be to determine how to provide and share this information with crowd control personnel, and the specific methods of crowd guidance are also future research challenges.

Acknowledgments

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This work was supported by JST's Mirai Program (JPMJMI20D1).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study may be available on reasonable request from the authors. The data are not publicly available due to privacy restrictions.

Endnotes

[1] In recent years, serious accidents have occurred, such as the Itaewon crowd crush in Seoul (2022), where 158 people died, and the Akashi fireworks footbridge incident (2001), which resulted in 11 deaths.

[2] Since pedestrian counting at L2 was straightforward, and measurement error was considered minimal, data collected by a single person was used.

[3] Walking speeds were measured using Tracker, an open-source video analysis tool (https://physlets.org/tracker/). Pedestrian trajectories were analyzed frame-by-frame every 0.1 s to compute average walking speed. Tracker is distributed under the GNU General Public License.

[4] Crosswalk 2 is about 5 m wide (striped portion), but in reality, pedestrians fan out wider than the marked crosswalk. Based on field observations we therefore calculated the flow coefficient using a 7 m width and also adopted 7 m in the simulation. The flow rate and flow-coefficient values for Crosswalk 2 are 5-min moving averages that include stoppage time, and the interval between red and blue signals follows the measured cycle of red 60 s/blue 80 s.

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[5] Density was estimated from still images captured every minute. Pedestrians who remained stopped between frames may have been included.

[6] Service Level (LOS) is often used in transport design. Levels D or above are acceptable; Level E or below often indicates restricted or dangerous flow requiring intervention.

[7] Refer Note 4.

[8] In addition to Pathfinder (Thunderhead Engineering), major crowd simulation software includes buildingEXODUS, STEPS, MassMotion, and SimTread.

[9] On June 4, the game remained close until the final inning (Table 5), resulting in few early departures. After the match, a large crowd exited at once, making this the most congested day.

[10] Agents were generated via the Occupant Sources component using random seeds, resulting in varied profiles and entry points.

[11] The representative simulation was the one with the fifth highest total NSE value among 10 trials.

[12] In the line L1_MA5, diverted flow was reduced. The segment from match end (0 min) to steady period end (43 min) was set at 65% of the actual L1 value.

[13] Calculated from the ratio of the number of people passing through L4 and L2 during the period t_{0-ec}.

[14] Of those who passed through L1, 71% were assumed to take Route 1, 26% Route 3 (adjusted from 29%), and 3% were assumed to split from Route 3 to L4.

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